# Listening Homework 2

Book 3 Unit 6

## Questions 1-5 Sentence completion

Write **ONE WORD, ONE NUMBER, OR ONE OF EACH** from the indicated SHORT audio. Do not add % where it is printed.

**1.** [6.1-S02] **Visitors can see a _____ of the fountain pen in Room 11B.**

**2.** [6.1-S03] **The same room contains early _____ of spectacles.**

**3.** [6.1-S03] **Spectacles were invented in _____ .**

**4.** [6.1-S03] **No one _____ who invented spectacles.**

**5.** [6.1-S03] **The first spectacles could be _____ on the nose.**

# Listening Homework 2

Book 3 Unit 6

## Questions 6-10 Sentence completion

Write **ONE WORD, ONE NUMBER, OR ONE OF EACH** from the indicated SHORT audio. Do not add % where it is printed.

**6.** [6.1-S03] **Spectacles helped people with poor _____ to read.**

**7.** [6.1-S04] **Al-Jazari lived in twelfth-century _____.**

**8.** [6.1-S04] **Al-Jazari's book lists one _____ different machines.**

**9.** [6.1-S05] **Water and _____ make the mechanical clocks work.**

**10.** [6.1-S05] **The crankshaft lets machines move in a _____ line.**

## Questions 11-15 Single choice

Tick **ONE** answer, A-C, for each question. Use the indicated SHORT audio.

**11.** [6.1-S01] **What is the title of the exhibition?**

- ☐ A. The Future of Ocean Travel
- ☐ B. A Guide to Cooking
- ☐ C. Discovering Medieval Science

**12.** [6.1-S01] **What name have the Middle Ages often been given?**

- ☐ A. The Dark Ages
- ☐ B. The Digital Ages
- ☐ C. The Football Ages

**13.** [6.1-S01] **What does the guide say about that traditional label for the Middle Ages?**

- ☐ A. The tour will confirm it is accurate
- ☐ B. The tour will show it is misleading
- ☐ C. The tour will discuss a railway timetable

**14.** [6.1-S01] **Which regions are named as sources of inventions brought to Europe?**

- ☐ A. Only Europe itself
- ☐ B. The Moon and Mars
- ☐ C. India and China among other regions

**15.** [6.1-S01] **What contribution does the guide attribute to medieval inventions?**

- ☐ A. They made a great contribution to society
- ☐ B. They had no effect on society
- ☐ C. They set the menu in the museum cafe

## Listening Homework 2

Book 3 Unit 6

### Questions 16-20 Single choice

Tick **ONE** answer, A-C, for each question. Use the indicated SHORT audio.

**16.** [6.1-S01] **What happened to many medieval inventions after their time?**

- ☐ A. All were immediately forgotten
- ☐ B. They became museum tickets
- ☐ C. They remain in use today

**17.** [6.1-S01] **How does the guide describe some medieval inventions today?**

- ☐ A. They are very common
- ☐ B. They have all disappeared
- ☐ C. They are listed on a lunch menu

**18.** [6.1-S07] **The scientists who invented gunpowder had an original aim. Did they achieve it?**

- ☐ A. Yes, they achieved it completely
- ☐ B. The guide discusses a train timetable instead
- ☐ C. No, they did not

**19.** [6.1-S07] **Who is named in connection with the first written gunpowder instructions?**

- ☐ A. Al-Jazari alone
- ☐ B. Zeng Gongliang, Ding Du and Yang Weide
- ☐ C. Three professional tennis players

**20.** [6.1-S02] **What problem did the first fountain pen help avoid?**

- ☐ A. Difficulty reading small letters
- ☐ B. Ink stains on fingers and clothes
- ☐ C. Food spoiling in a refrigerator

## Questions 21-23 Multiple choice

Tick **TWO** answers, A-E, for each numbered question. Use the indicated SHORT audio.

**21.** [6.1-S06] **Which TWO details locate the invention of gunpowder?**

- ☐ A. It was found on the Moon.
- ☐ B. It arose in Europe.
- ☐ C. It dates from the ninth century.
- ☐ D. It dates from the nineteenth century.
- ☐ E. It arose in China.

**22.** [6.1-S06] **Which TWO details does the guide give about the invention of gunpowder?**

- ☐ A. The scientists aimed to improve engines.
- ☐ B. The scientists aimed to extend life.
- ☐ C. The guide says its culture had only two great inventions.
- ☐ D. The guide calls it one of four great inventions from its culture.
- ☐ E. The project selected a football captain.

**23.** [6.1-S07] **Which TWO details does the guide give about the experiments and the first written gunpowder instructions?**

- ☐ A. The instructions were written in the twenty-first century.
- ☐ B. The instructions were written in the eleventh century.
- ☐ C. The experiments produced only reading glasses.
- ☐ D. The instructions include a museum café menu.
- ☐ E. The experiments led to fireworks and weapons.

## Questions 24-25 Multiple choice

Tick **TWO** answers, A-E, for each numbered question. Use the indicated SHORT audio.

**24.** [6.1-S08] **Which TWO historical effects of gunpowder are stated?**

- ☐ A. It ended every medieval battle immediately.
- ☐ B. It affected only one local exhibition.
- ☐ C. It altered the results of many medieval battles.
- ☐ D. It affected the history of the world.
- ☐ E. It improved the flavour of bread.

**25.** [6.1-S08] **Which TWO points close this part of the tour?**

- ☐ A. Many people regard gunpowder as history's most important invention.
- ☐ B. Everyone agrees that gunpowder is unimportant.
- ☐ C. The group will return to the fountain-pen display.
- ☐ D. The group will judge a dance contest.
- ☐ E. The group will continue to a display about medieval medicine.

## Passage B - From Shark Riblets to Aircraft Film

**A** Shark skin offers a different lesson in biomimicry. Its useful feature is not a shark-like **appearance** but a surface covered with tiny, aligned ribs called riblets. Water moving past the animal meets this repeated structure. Engineers have investigated whether a related pattern can reduce friction on vehicles that travel through a fluid. For aircraft, the fluid is air, so the natural model has to be adapted rather than copied as skin.

**B** Frictional drag develops where air meets an aircraft's outer surface. Riblets guide the near-surface air in the direction of travel and can reduce part of that resistance under suitable **conditions**. The pattern does not push an aircraft forward. It changes the **movement** of air very close to the surface. Because the ribs are microscopic, their size, direction and position matter to the result. A random rough surface would not provide the same controlled geometry.

**C** Lufthansa Technik and BASF worked to **translate** this principle into AeroSHARK, a film carrying microscopic riblets. Correct alignment is **essential** because the grooves must follow the local airflow. The film also has to fit the curved fuselage and engine covers while meeting maintenance and aviation requirements. The finished technology therefore combines a biological idea with materials engineering, installation planning and certification. It is not a sheet of real shark skin.

## Passage B - From Shark Riblets to Aircraft Film

**D** On 14 October 2022, SWISS began passenger operations with a Boeing 777-300ER carrying about 950 square metres of the film on its fuselage and engine nacelles. Lufthansa Technik said simulations calculated fuel and emission savings of just over one percent for that aircraft type. The scheduled flights were intended to test the estimate in daily operation. That sequence prevents a forecast from being reported as a measured outcome. These details **illustrate** the difference between a calculated potential and a result still being validated.

**E** In December 2022, the European Union Aviation Safety Agency granted a supplemental type certificate for the 777-300ER and 777F modifications. Certification allowed series installation on those two variants. Approval also concerns a defined installation, not the entire biological idea. It did not approve an **unlimited range** of aircraft. The developers reported that using less fuel would also reduce carbon dioxide, a **harmful** greenhouse gas. The size of the benefit still depends on aircraft type, covered area, routes and operating patterns.

**F** AeroSHARK shows why engineers must **rethink** a natural structure for a new environment. The riblets provide a design principle, but testing establishes where the film works and how large a saving is reasonable to claim. The 2022 sources distinguish simulation, daily-flight validation and regulatory approval rather than treating them as one kind of evidence. That distinction helps readers judge the technology without turning a one-percent estimate into a promise for every aircraft or every journey.

## Reading Homework 2

Book 3 Unit 6

### Questions 1-5 True, False, Not Given

TRUE: agrees with the passage. FALSE: contradicts it. NOT GIVEN: insufficient information. Tick **ONE** box.

**1. The useful shark-skin feature is its shark-like appearance.**

☐ TRUE ☐ FALSE ☐ NOT GIVEN

**2. The engineers used the great white shark as their specific research model.**

☐ TRUE ☐ FALSE ☐ NOT GIVEN

**3. The riblets are arranged in aligned rows.**

☐ TRUE ☐ FALSE ☐ NOT GIVEN

**4. Any randomly rough aircraft surface would provide the same geometry as riblets.**

☐ TRUE ☐ FALSE ☐ NOT GIVEN

**5. AeroSHARK is made from sheets of real shark skin.**

☐ TRUE ☐ FALSE ☐ NOT GIVEN

## Questions 6-10 True, False, Not Given

TRUE: agrees with the passage. FALSE: contradicts it. NOT GIVEN: insufficient information. Tick **ONE** box.

**6. The first SWISS passenger aircraft described carried about 950 square metres of film.**

☐ TRUE ☐ FALSE ☐ NOT GIVEN

**7. The reported simulation estimate was exactly ten percent for that aircraft type.**

☐ TRUE ☐ FALSE ☐ NOT GIVEN

**8. Certification allowed series installation on the two approved aircraft variants.**

☐ TRUE ☐ FALSE ☐ NOT GIVEN

**9. EASA required every certified aircraft to fly the same route.**

☐ TRUE ☐ FALSE ☐ NOT GIVEN

**10. The film needs replacing after exactly five years.**

☐ TRUE ☐ FALSE ☐ NOT GIVEN

## Questions 11-15 Matching information

Which paragraph contains each piece of information? Tick **ONE** box. You may use a letter more than once.

**11. The engineered product combines a biological idea with materials and installation planning.**

☐ A ☐ B ☐ C ☐ D ☐ E ☐ F

**12. The film must fit curved aircraft parts while meeting maintenance requirements.**

☐ A ☐ B ☐ C ☐ D ☐ E ☐ F

**13. The benefit can vary with routes and operating patterns.**

☐ A ☐ B ☐ C ☐ D ☐ E ☐ F

**14. Air is identified as the fluid around an aircraft.**

☐ A ☐ B ☐ C ☐ D ☐ E ☐ F

**15. The size, direction and position of microscopic ribs affect the result.**

☐ A ☐ B ☐ C ☐ D ☐ E ☐ F

## Questions 16-20 Matching information

Which paragraph contains each piece of information? Tick **ONE** box. You may use a letter more than once.

**16. Simulation, daily-flight validation and approval are treated as different evidence.**

☐ A ☐ B ☐ C ☐ D ☐ E ☐ F

**17. A repeated surface pattern is investigated as a way to limit vehicle friction.**

☐ A ☐ B ☐ C ☐ D ☐ E ☐ F

**18. A forecast is deliberately separated from a result still being tested.**

☐ A ☐ B ☐ C ☐ D ☐ E ☐ F

**19. The size of the benefit can depend on the area covered.**

☐ A ☐ B ☐ C ☐ D ☐ E ☐ F

**20. Passenger operations began on a dated aircraft carrying a stated covered area.**

☐ A ☐ B ☐ C ☐ D ☐ E ☐ F

## Questions 21-23 Multiple choice

Tick **TWO** answers, A-E, for each numbered question.

**21. Which TWO statements describe the natural model in Paragraph A?**

- ☐ A. The model explains how a piano is tuned.
- ☐ B. Water moves past the animal's surface.
- ☐ C. The pattern is designed to stop all water movement.
- ☐ D. The model involves a repeated surface structure.
- ☐ E. The surrounding fluid around the shark is air.

**22. Which TWO statements explain how the engineered riblet film is intended to work?**

- ☐ A. It guides air close to the surface in the travel direction.
- ☐ B. It keeps food cold in a refrigerator.
- ☐ C. Its grooves must follow the local airflow.
- ☐ D. It pushes the aircraft forward by itself.
- ☐ E. Its direction has no effect on performance.

**23. Which TWO aircraft parts carried the film in the SWISS example?**

- ☐ A. The fuselage
- ☐ B. The whole of both wings
- ☐ C. The landing gear
- ☐ D. The shelves of an ocean clinic
- ☐ E. The engine nacelles

## Questions 24-25 Multiple choice

Tick **TWO** answers, A-E, for each numbered question.

**24. Which TWO points describe the reported environmental benefit?**

- ☐ A. Carbon dioxide is described as harmless
- ☐ B. Using less fuel would reduce carbon dioxide
- ☐ C. Approval set the opening hours of a library
- ☐ D. Using less fuel would increase carbon dioxide
- ☐ E. Carbon dioxide is described as a harmful greenhouse gas

**25. Which TWO questions does testing help engineers answer?**

- ☐ A. Whether testing can be skipped after finding a natural pattern
- ☐ B. Whether one result guarantees equal savings on every aircraft
- ☐ C. Where the film works
- ☐ D. How large a saving is reasonable to claim
- ☐ E. When birds migrate between continents

# Speaking Homework 2

Book 3 Unit 6

## Questions 1-5 Build sentences

Put the word groups in order. Use each group once. Do not add words or change word forms. Record each full sentence.

**1. Start with "The improved version".**
a different room, The improved version, was tested, in

**2. Start with "These small parts".**
recycled plastic, from, These small parts, are made

**3. Start with "The new label".**
the instructions, clearer, The new label, makes

**4. Start with "Several changes".**
after, were made, Several changes, the first test

**5. Start with "The device".**
by, The device, is used, students with different needs

## Questions 6-10 Complete the replies

Read each situation and follow its clue to complete the reply. Record the whole reply.

**6. Situation:** Two classmates tested the new version. Choose was or were.

**Friend: Did anyone try the new design?**

You: Yes, it _____ tested by two classmates.

**7. Situation:** The new stand is easier to fold. Choose easily or easier.

**Friend: What does the new joint do?**

You: It makes the stand _____ to fold.

**8. Situation:** The designers changed three parts in response to the feedback. Choose made or taken.

**Friend: Was the feedback useful?**

You: Yes, three changes were _____ in response to it.

**9. Situation:** The cards contain short written instructions. Choose use or used.

**Friend: What are these cards for?**

You: They are _____ to explain each step.

**10. Situation:** Plastic is the material. Choose from or by.

**Friend: What are the small parts made from?**

You: They are made _____ recycled plastic.

# Speaking Homework 2

Book 3 Unit 6

## Questions 11-15 Order and speak

Put sentences A-C in order. Record the opening, then read all three sentences in the correct order. Do not read the letters.

**11. Explain a redesign after testing.**

**Opening:** The first test gave us a useful surprise.

| | |
|---|---|
| A | It was replaced with a larger handle. |
| B | The larger version made the object easier to lift. |
| C | The original handle was difficult to hold. |

**12. Explain how a prototype is shared.**

**Opening:** Other students need to understand our idea.

| | |
|---|---|
| A | Those photographs were added to a short guide. |
| B | The prototype was photographed from three sides. |
| C | The guide was shared with the next test group. |

**13. Describe a choice between materials.**

**Opening:** We tried two materials before deciding.

| | |
|---|---|
| A | Because of that tearing, a stronger card cover was used for the next version. |
| B | That paper tore during repeated use. |
| C | The first cover was made from thin paper. |

**14. Explain an accessibility improvement.**

**Opening:** We wanted more people to read the controls.

| | |
|---|---|
| A | The change made the controls easier to read. |
| B | The old buttons had very small labels. |
| C | Those labels were replaced with larger ones. |

**15. Explain how feedback changed a project.**

**Opening:** We did not notice the problem until someone else tried it.

| | |
|---|---|
| A | That hinge was adjusted before the second test. |
| B | She found that one hinge was too stiff. |
| C | A classmate tested the folding stand. |

# Speaking Homework 2

Book 3 Unit 6

## Questions 16-18 Follow the model

Use the model to help you express different content in 4 connected sentences. Write key words, then record your answer.

**16. Describe a first design, feedback and a change. Use two passive forms.**

**Model:** Our first model was made from thin card. It was tested by another group. They found that the sides bent too easily. Stronger card was used for the next model.

**17. Explain how a change makes an invention easier for a new user.**

**Model:** The instructions were originally printed in small text. Larger words were added after a test. The change makes each step easier to read. New users can follow the guide more confidently.

**18. Describe how a device would be introduced to a test group.**

**Model:** The device would first be placed on a table. Each student would be shown its main button. A short task would then be explained. Their comments could guide the next design.

# Speaking Homework 2

Book 3 Unit 6

## Questions 19-20 Follow the model

Use the model to help you express different content in 4 connected sentences. Write key words, then record your answer.

**19. Compare two versions of a familiar object using passive forms and make.**

**Model:** The first bag was made without a side pocket. A pocket was added to the new version. It makes small items easier to reach. The main section can still hold larger objects.

**20. Describe a classroom invention from materials to use. Include a specific benefit.**

**Model:** The model organizer was made from old boxes. The boxes were fixed inside a larger tray. It is used to separate small parts. This makes preparing for a project quicker.

# Speaking Homework 2

Book 3 Unit 6

## Questions 21-22 Describe the picture

Write key words, then record 3 connected sentences for each question. Use the words if helpful.



Words: device, install, outcome

**21. Describe the bicycle light and the work being done on it. Use a passive form.**

Words: make improvements, user, create

**22. Imagine you designed this light. Describe one improvement and a test of it.**

# Speaking Homework 2

Book 3 Unit 6

## Questions 23-25 Describe the picture

Write key words, then record 3 connected sentences for each question. Use the words if helpful.



Words: create, product, make a contribution

**23. Describe the wooden object and how the two people are making it.**

Words: access, allow, product

**24. Suggest a use for the organizer in a shared workshop. Explain how it could help.**

Words: make improvements, user, make a decision

**25. Imagine the makers want to improve their organizer. Suggest a change and explain how to choose it.**

## Writing Homework 2

Book 3 Unit 6

### Questions 1-5 Build sentences

Put the word groups in order. Use each group once. Do not add words or change word forms. Write each full sentence.

**1. Start with "The redesigned handle".**

be, will probably, The redesigned handle, easier to hold

**2. Start with "The new feature".**

may not, work, on every device, The new feature

**3. Start with "We should".**

rethink, of the instructions, the order, We should

**4. Start with "A clearer display".**

A clearer display, to follow the steps, could enable, more users

**5. Start with "The unfinished model".**

is unlikely, to be ready, The unfinished model, tomorrow

### Questions 6-10 Complete the sentences

Complete each sentence. Follow the word choices or clues in each question.

**6. Choose probable or probably.**

The redesigned stand will _____ be lighter.

**7. Add re- to think to mean think again.**

We may need to _____ the plan.

**8. Choose may or must. The result is possible but uncertain.**

The new version _____ reduce the waiting time.

**9. Choose enabled or enable after could.**

Larger controls could _____ more people to use it.

**10. Add dis- to agree to mean hold a different opinion.**

Two designers may _____ about the best shape.

## Writing Homework 2

Book 3 Unit 6

### Questions 11-15 Choose the sentence order

Read sentences A-C. Tick one sequence for each question.

**11. Order a prediction after a design change.**

A A winter test could check this prediction.
B The team added a larger handle to the bag.
C That handle may be easier to hold with gloves.

☐ A-C-B ☐ C-A-B ☐ B-C-A

**12. Order a feature proposal.**

A The team should ask those users to try a sample.
B The app could include spoken instructions.
C Those instructions might help users who find small text difficult.

☐ B-C-A ☐ C-A-B ☐ B-A-C

**13. Order an account of rethinking a design.**

A One way to rethink those materials is to use a lighter one to solve the weight problem.
B To address that weight problem, the designer decided to rethink its materials.
C The first case was too heavy to carry comfortably.

☐ B-C-A ☐ C-B-A ☐ B-A-C

**14. Order a limitation and alternative.**

A Some visitors may arrive without one.
B The digital guide needs a charged device.
C A printed version could provide an alternative for them.

☐ B-C-A ☐ A-C-B ☐ B-A-C

**15. Order a prediction based on limited testing.**

A Longer use might reveal different problems.
B The new frame worked in one short test.
C The group should therefore test it again before making a firm prediction.

☐ A-C-B ☐ C-B-A ☐ B-A-C

### Questions 16-18 Follow the model

Use the model to help you express different content in 4 connected sentences. Write your own answer with different details.

**16. Predict how a redesigned object may help a different user. Include a limitation and a test.**

**Model:** A bag with a larger handle may help someone wearing gloves. The handle would be easier to grip. However, it might catch on nearby objects. A short test could compare both problems.

**17. Describe a feature you would add to an app and predict its effect cautiously.**

**Model:** The app could add a spoken reminder. Users might notice it without looking at the screen. However, some people may prefer a quiet alert. A choice of settings could meet different preferences.

**18. Explain why a model needs rebuilding and what may improve.**

**Model:** The first model bent when a small weight was added. The team should rebuild its centre section. A stronger joint may support the weight better. Another test would show whether the change works.

# Writing Homework 2

Book 3 Unit 6

## Questions 19-20 Follow the model

Use the model to help you express different content in 4 connected sentences. Write your own answer with different details.

**19. Make two predictions about a classroom invention and include uncertainty.**

**Model:** A shared display could make instructions easier to see. It may also reduce the need for printed sheets. However, users at the back might struggle with small text. The teacher could test it from several seats.

**20. Describe a disagreement between designers and a way to compare their ideas.**

**Model:** One designer believes the tool should be very small. Another thinks larger buttons would enable easier use. Both ideas may have benefits. Testing two versions could help them compare the results.

# Writing Homework 2

Book 3 Unit 6

## Questions 21-22 Describe the picture

Write 3 connected sentences for each question. Use the words if helpful.



Words: personal, image, movement

**21. Describe the traveller and the phone. Make a cautious prediction about the next few minutes.**

Words: translate, rethink, essential

**22. Imagine the traveller cannot understand a route message. Predict how an app could help and mention a limitation.**

# Writing Homework 2

Book 3 Unit 6

## Questions 23-25 Describe the picture

Write 3 connected sentences for each question. Use the words if helpful.



Words: movement, illustrate, conditions

**23. Describe the model and the fan. Predict what may happen when the fan is switched off.**

Words: conditions, movement, illustrate

**24. Imagine the students move the fan farther away. Predict an outcome and suggest what they should observe.**

Words: preview, unsafe, essential

**25. Imagine the students will show the model to younger pupils. Suggest how to prepare the demonstration.**